Cloudflare's input

for

CMS

Request for Input on Improving Technology to Empower Medicare Beneficiaries
CMS-0042-NC

6/16/25

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June 16, 2025

Centers for Medicare & Medicaid Services (CMS)
7500 Security Boulevard
Baltimore, MD 21244

CMS,

We at Cloudflare are grateful for CMS' public outreach effort and are compelled to respond to CMS' Request for Public Input on Improving Technology to Empower Medicare Beneficiaries.

At Cloudflare, we have our eyes set on an ambitious goal — to help build a better Internet. We believe that with our talented team, smart technology and engaged users we can solve some of the biggest problems on the Internet.

Today, everything needs to be connected to everything, everywhere, all the time. This hyperconnectivity creates new challenges related to security, performance, resilience, and privacy.

As the world's first connectivity cloud, Cloudflare helps connect and protect millions of customers globally. Everyone from individuals to the world's largest enterprises use our unified platform of networking, security, and developer services to succeed in the everywhere world.

We are excited to offer thought leadership comments for CMS to review and appreciate the opportunity to engage in a future discussion if CMS desires.

Please contact me with any questions about Cloudflare's proposal.

Thank you for your consideration.

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Executive Summary

Cloudflare, Inc. is pleased to submit its response to the Centers for Medicare & Medicaid Services (CMS) and the Assistant Secretary for Technology Policy (ASTP)/Office of the National Coordinator for Health Information Technology (ONC) Request for Information (RFI) on the "Health Technology Ecosystem." This response focuses specifically on Section E, "Technology Vendors, Data Providers, and Networks," outlining how Cloudflare's comprehensive suite of security, performance, and reliability solutions directly addresses the RFI's inquiries and supports the advancement of a secure, interoperable, and patient-centric digital health ecosystem.

Cloudflare's global network and integrated platform are uniquely positioned to provide the foundational infrastructure necessary for digital health products to thrive, ensuring sensitive health information is protected, applications are highly performant, and access is streamlined for all stakeholders. Our commitment to regulatory compliance, including HIPAA and HITECH, underpins all capabilities, enabling confidence in the secure exchange of Protected Health Information (PHI).

About Cloudflare

Our History

Cloudflare was founded on July 17, 2009 by Matthew Prince, Michelle Zatlyn, and Lee Holloway as a project focused on improving website security and performance. Initially emerging from a Harvard Business School business plan competition, the company quickly gained attention for its approach to combating online threats. In 2010, Cloudflare launched its first product at TechCrunch — a service that not only protected websites from DDoS attacks, but also optimized web traffic through a content delivery network (CDN). Since then, Cloudflare has launched dozens of products and hundreds of features, opened offices across multiple countries, and created one of the world's largest and most interconnected networks that blocks billions of threats online for its customers every day.

As the leading connectivity cloud company, Cloudflare empowers organizations to make their employees, applications and networks faster and more secure everywhere, while reducing complexity and cost. Cloudflare's connectivity cloud delivers the most full-featured, unified platform of cloud-native products and developer tools, so any organization can gain the control they need to work, develop, and accelerate their business. It is trusted by millions of organizations – from the largest brands to entrepreneurs and small businesses to nonprofits, humanitarian groups, and governments across the globe.

Learn more about Cloudflare's connectivity cloud at cloudflare.com/connectivity-cloud. Learn more about the latest Internet trends and insights at radar.cloudflare.com.

Our Mission

Cloudflare's mission – to help build a better internet – is rooted in the belief that the internet can be a powerful tool for positive change. To achieve this goal, we focus on:

- making the Internet more accessible by improving reliability around the world

- engineering privacy protections into the basic infrastructure of the Internet
- designing a network that helps band businesses, people, and organizations together to be stronger than malicious actors online, and
- building together as part of a community that is striving to make the Internet a force for good

We are committed to improving internet security, performance, and reliability for individuals, businesses, and public institutions around the world and proudly report our progress on our various initiatives through our Cloudflare Impact page at: cloudflare.com/impact/

Our Network

Cloudflare has built an efficient, scalable network that allows us to rapidly develop and deploy our products for our customers and that is architected to be flexible, scalable, and get more and more efficient as it expands.

Our network is designed to be able to grow capacity quickly and inexpensively; to allow for every server, in every city, to run nearly every Cloudflare service; and to allow us to shift customers and traffic across our network efficiently. We refer to this architecture as "serverless" because it means we can deploy standard, commodity hardware, and our product developers and customers do not need to worry about the underlying servers. Our software is designed to manage the deployment and execution of our product developers' code and our customers' code across our network. Because we manage the execution and prioritization of code running across our network, it means that we are both able to improve the performance of our highest paying customers, and also effectively leverage idle capacity across our network. We have chosen to utilize this idle capacity to create a free tier of service which has generated substantial global scale for us. In turn, this scale makes us attractive partners for Internet Service Providers (ISPs) globally, which reduces our co-location and bandwidth costs. As our network grows, these dynamics become even more powerful.

Today, our network spans more than 335 cities in over 125 countries worldwide, offers 348 Tbps of capacity, and interconnects with over 13,000 networks globally, including major ISPs, cloud services, and enterprises.

Cloudflare's Global Network (as of Q4 2024)

Other Facts and Stats

- **Type of Business:** For profit, publicly traded C Corporation
- **Incorporation:** July 17, 2009 in Delaware
- **Corporate Headquarters Address:** 101 Townsend Street, San Francisco, CA, 94107, USA
- **Exchange/Ticker Symbol:** New York Stock Exchange (NYSE). Our Ticker Symbol is NET.
- **Office Locations:** 15 (San Francisco, CA, Austin, TX, Champaign, IL, Washington, D.C., Mexico City, Dubai, Lisbon, London, Munich, Paris, Beijing, Seoul, Singapore, Sydney, and Tokyo)
- **Total Employees:** More than 4,200 people

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Cloudflare Response

Cloudflare Response to Centers for Medicare & Medicaid Services (CMS) Request for Information (RFI) on "Health Technology Ecosystem"

Section E. Technology Vendors, Data Providers, and Networks

TD-1: What are the most effective ways for CMS/ASTP/ONC to stimulate developer interest in building digital health products for Medicare beneficiaries?

Cloudflare believes that stimulating developer interest requires reducing the inherent complexities and risks associated with building and deploying digital health products, particularly concerning security and compliance. Our platform achieves this by:

- **Enabling Focus on Innovation:** Cloudflare's comprehensive suite of security, performance, and reliability services abstracts away significant infrastructure management, security patching, and compliance overhead. This allows developers to dedicate their resources and creativity to core product features and innovation, accelerating the development of impactful digital health solutions.[1]
- **Simplifying Security and Compliance:** Cloudflare's API Shield automatically discovers API endpoints, blocks common API security risks (including OWASP Top 10), prevents data exfiltration, and validates schemas, ensuring that APIs handling sensitive health data are secure by design without requiring extensive security expertise from every developer.[3] This reduces the "security tax" on innovation.
- **Accelerate Development and Deployment:** Cloudflare's developer-friendly solutions streamline the development and deployment lifecycle. This proactive security approach, by reducing common security and operational hurdles, translates to faster time-to-market for new digital health products.[5]
- **Improve Efficiency and Reduce Costs:** Independent analysis has shown that Cloudflare's connectivity cloud delivers a substantial Return on Investment (ROI) over three years by boosting productivity, enhancing security efficiency, and reducing the likelihood and costs associated with data breaches. This financial benefit makes investment in digital health product development more attractive and sustainable.[6]

TD-2: Regarding CMS Data, to stimulate developer interest—

a. What additional data would be most valuable if made available through CMS APIs?

While Cloudflare does not provide the data itself, our expertise lies in enabling the secure and performant exposure of any data via APIs. To stimulate innovation, additional data that would be most valuable if made available through CMS APIs includes:

- **Comprehensive Claims and Encounter Data:** Granular, de-identified claims and encounter data beyond what is currently available, including detailed service codes, diagnoses, and procedure information, could enable developers to build more sophisticated analytics tools for population health management, care coordination, and value-based care initiatives.
- **Social Determinants of Health (SDOH) Data:** Integrating de-identified SDOH data (e.g., housing stability, food security, transportation access, education levels) with clinical data could allow

developers to create holistic digital health products that address non-medical factors impacting health outcomes.

- **Provider Directory with Enhanced Attributes:** A more comprehensive and frequently updated provider directory, including specialties, languages spoken, accessibility features, and participation in specific Alternative Payment Models (APMs), would enable more precise care navigation applications.
- **Quality Measure Data:** More detailed, real-time quality measure data, potentially aligned with FHIR-based eCQMs, could allow developers to build tools that provide actionable insights for providers to improve care quality and for beneficiaries to make informed decisions about their care.[7]

b. What data sources are most valuable alongside the data available through the Blue Button 2.0 API?

Alongside the Blue Button 2.0 API, which provides Medicare claims and benefit information [1], the most valuable data sources for developers would include:

- **Electronic Health Record (EHR) Data:** Access to a broader range of clinical data from EHRs, including progress notes, lab results, imaging reports, and medication lists, would provide a more complete picture of a patient's health journey. FHIR-based APIs are crucial for standardizing this exchange.[7]
- **Consumer-Generated Health Data:** Data from wearables, remote monitoring devices, and patient-reported outcomes (PROs) can offer real-time insights into a beneficiary's daily health and lifestyle, enabling personalized health management applications.
- **Public Health Data:** Integration with public health registries, immunization records, and syndromic surveillance data could support applications for disease prevention, outbreak management, and community health initiatives.
- **Payer-Specific Data:** Beyond Medicare claims, access to de-identified data from other payers (Medicaid, commercial insurers) could provide a broader understanding of healthcare utilization and costs across different populations.

c. What obstacles prevent accessing these data sources today?

Obstacles preventing access to these valuable data sources today primarily revolve around security, interoperability, and operational complexities:

- **Security and Privacy Concerns:** The highly sensitive nature of Protected Health Information (PHI) leads to significant apprehension among data holders regarding unauthorized access, data breaches, and compliance violations (e.g., HIPAA). This often results in a cautious approach to exposing data via APIs.[5] Cloudflare mitigates this by providing robust API security, DDoS protection, and data loss prevention, creating a secure environment for data exposure.[6]
- **Lack of Standardized APIs and Data Formats:** Inconsistent data formats and the absence of widely adopted, standardized APIs (beyond current FHIR implementations) make it challenging for developers to integrate disparate data sources efficiently.[7]
- **Technical Complexity and Cost of API Development/Maintenance:** Building, securing, and maintaining robust APIs that can handle high volumes of requests and ensure low latency is technically complex and costly for many organizations. This includes managing API discovery, authentication, schema validation, and threat mitigation.[3]
- **Information Blocking Practices:** While often unintentional, technical limitations, security concerns, or

a lack of incentives can lead to practices that impede the seamless flow of health information.[1] Cloudflare addresses technical barriers by ensuring secure and performant data exchange, reducing the technical rationale for information blocking.[10]

- **Identity Management Fragmentation:** The absence of a unified, secure digital identity framework across the ecosystem complicates authentication and authorization for accessing diverse data sources.[1] Cloudflare's identity-agnostic Zero Trust platform simplifies this by integrating with various IdPs.[13]

d. What other APIs should CMS and ASTP/ONC consider including in program policies to unleash innovation and support patients and providers?

To unleash innovation and support patients and providers, CMS and ASTP/ONC should consider including program policies for APIs that facilitate administrative, operational, and real-time data exchange:

- **Administrative Workflow APIs:** APIs for appointment scheduling, referral management, prior authorization status, and patient intake forms could significantly reduce administrative burden and improve efficiency for both patients and providers.[1]
- **Real-time Clinical Event APIs:** APIs that provide real-time alerts for critical clinical events (e.g., hospital admissions/discharges, emergency department visits, significant lab results) could enable proactive care coordination and intervention.
- **Provider Credentialing and Enrollment APIs:** Standardized APIs for verifying provider credentials, licensing, and enrollment status could streamline administrative processes for payers and providers.
- **Public Health Reporting APIs:** APIs to facilitate automated, secure reporting of public health data (e.g., communicable diseases, syndromic surveillance) to relevant authorities, reducing manual burden and improving public health response.
- **AI/ML Model Interaction APIs:** As AI becomes more prevalent, APIs that allow secure and governed interaction with AI/ML models for clinical decision support, predictive analytics, and personalized health recommendations could unleash significant innovation.[16] Cloudflare's emerging Model Context Protocol (MCP) concept aligns with this, aiming to enable seamless AI system connectivity with existing business tools and data.[16]

TD-3: What are the impacts (positive and negative) of implementing digital identity credentials for patients, caregivers, providers, payers, and technology vendors?

Cloudflare's identity-agnostic Zero Trust platform is specifically designed to maximize the positive impacts and mitigate the negative ones associated with digital identity credentials across the entire health ecosystem:

Positive Impacts (Enabled by Cloudflare):

- **Simplified, Secure Access for Patients & Caregivers:** Cloudflare's Zero Trust integrates seamlessly with a wide array of identity providers (IdPs), including social logins and open standards like SAML and OIDC. This allows patients to use familiar methods for secure, streamlined access to multiple digital health products and portals without the burden of managing numerous separate accounts, thereby reducing login fatigue and improving engagement.[12]
- **Enhanced Clinician Productivity:** By replacing cumbersome Virtual Private Networks (VPNs) with

Zero Trust Network Access (ZTNA), Cloudflare provides fast, reliable, and secure access to Electronic Health Records (EHRs) and internal systems from any location. This significantly improves provider workflows, reduces IT burden, and allows clinicians to focus more on patient care.[6]

- **Secure and Verifiable Data Exchange for Payers:** Cloudflare's API security solutions offer robust authentication mechanisms (e.g., mTLS certificates, JSON Web Tokens (JWT), API keys, OAuth 2.0 tokens) for API access. This ensures that only authorized applications and systems can exchange sensitive health data, which is crucial for accelerating interoperability mandates and supporting value-based care models.[3]
- **Reduced Security Overhead for Technology Vendors:** Cloudflare simplifies API security for developers, fostering innovation by allowing them to build and integrate new digital health products without extensive security complexities.[3]
- **Foundation for Value-Based Care (VBC):** Consistent and secure identity verification across disparate data sources and applications provides the necessary foundation for data-driven VBC models, enabling proactive interventions and improved care coordination through reliable data access.[15]

Negative Impacts (Mitigated by Cloudflare):

- **Integration Complexity:** Cloudflare's identity-agnostic approach allows seamless integration with leading corporate IdPs (e.g., Okta, Azure AD), social identities, and open standards, eliminating the need for complex custom integrations that often create security gaps and operational overhead.[13]
- **Security Risks (e.g., "Blast Radius" of Breaches):** Zero Trust architectures provide granular access controls, ensuring users only access the specific data and systems they are authorized for. This significantly reduces the "blast radius" of a potential data breach, as compromised credentials would have limited access.[12]
- **User Friction:** Cloudflare facilitates Single Sign-On (SSO) and Multi-Factor Authentication (MFA) across applications, eliminating unnecessary login steps and integrating strong authentication methods to improve user experience while simultaneously enhancing the security posture.[12]

TD-4: How can CMS/ASTP/ONC encourage the use of open standards-based APIs (e.g., FHIR) for health data exchange among technology vendors, data providers, and networks?

CMS/ASTP/ONC can encourage widespread FHIR adoption by promoting solutions that make these APIs inherently secure, reliable, and easy to manage. Cloudflare directly supports this by:

- **Securing FHIR APIs:** Cloudflare's API Shield provides comprehensive protection against API-specific attacks, including automated discovery of all public API endpoints, schema validation to ensure only legitimate requests are processed, and robust built-in authentication mechanisms (mTLS, JWT, API keys, OAuth 2.0) to ensure only authorized clients access sensitive health data.[3] This directly addresses security concerns that might otherwise hinder FHIR adoption.
- **Ensuring Performance and Availability:** Cloudflare's global network and performance optimizations ensure that FHIR APIs are fast and continuously available, which is critical for real-time data exchange and clinical decision support.[3]
- **Mitigating Information Blocking:** By creating a secure and reliable environment for data exchange, Cloudflare helps remove technical barriers that could lead to information blocking, fostering greater confidence in opening up data access via FHIR APIs.[10]

TD-5: What are the benefits of a nationwide provider directory of FHIR endpoints for technology vendors, data providers, and networks?

A nationwide provider directory of FHIR endpoints would significantly enhance interoperability by centralizing discoverability. Cloudflare's role is to ensure the security and performance of these endpoints and the directory itself, thereby maximizing its benefits:

- **Secure Access to Endpoints:** Cloudflare's API Shield, Web Application Firewall (WAF), and DDoS protection safeguard the FHIR endpoints listed in such a directory from various cyber threats, including volumetric attacks, bot abuse, and malicious payload exploits. This ensures that vendors, data providers, and networks can access them securely and reliably.[23]
- **Accelerated Data Exchange:** Cloudflare's global network and Content Delivery Network (CDN) capabilities ensure fast and reliable access to these FHIR endpoints, accelerating data exchange across the nationwide ecosystem.[3]
- **Building Trust and Reliability:** By providing robust security and performance, Cloudflare helps build trust in the directory and the underlying FHIR endpoints, encouraging broader adoption and utilization by all stakeholders.[18]

TD-6: What unique functions, if any, do Qualified Health Information Networks (QHINs) or other data sharing networks perform that could not be achieved through point-to-point FHIR API connections?

QHINs and data sharing networks offer unique benefits beyond point-to-point connections, such as centralized trust, broader reach, and enhanced governance. Cloudflare's platform can serve as a foundational layer for these networks, enabling their unique functions:

- **Ensuring Network Reliability, Privacy, Security, and Trust:** Cloudflare's global network and comprehensive security services (Zero Trust, DDoS protection, WAF, API Shield) provide the underlying infrastructure to ensure the reliability, privacy, security, and trust required for large-scale data sharing networks like those envisioned by TEFCA™.[6]
- **Unified Policy Enforcement and Visibility:** Cloudflare's single control plane allows for consistent policy enforcement and provides granular visibility into traffic and access events across the entire network. This is critical for managing complex data sharing agreements and audit requirements within QHINs.[6]
- **Scalability and Performance:** Cloudflare's network is designed to handle massive traffic surges and optimize data routing, ensuring that data sharing networks remain performant and available even under high demand, as demonstrated by Verto Health's experience during the COVID-19 pandemic.[18]
- **Integration with AI and Analytics:** Cloudflare's ability to push granular log data to Security Information and Event Management (SIEM) tools like Splunk or Sumo Logic [15], combined with emerging concepts like the Model Context Protocol (MCP) [16], can facilitate the secure integration of AI systems and analytics across diverse data sources within a network, enabling more intelligent and efficient data utilization.

TD-7: What are the impacts (positive and negative) of the USCDI (United States Core Data for Interoperability) on technology vendors, data providers, and networks, and what are its limitations?

The USCDI standardizes health data, which is a significant positive step for interoperability. Cloudflare's impact is in ensuring the secure and efficient exchange of this standardized data:

- **Positive Impact (Enabled by Cloudflare):** Cloudflare's API security solutions (API Shield, WAF, DDoS protection) ensure that when USCDI-defined data is exchanged via APIs, it is protected from cyber threats and data leakage.[23] This allows technology vendors, data providers, and networks to confidently leverage the standardized data for innovation.
- **Enhanced Performance for Data Exchange:** Cloudflare's global network accelerates the transmission of USCDI-compliant data, ensuring fast and reliable interoperability across the ecosystem.[3]
- **Mitigating Limitations:** While USCDI is crucial for standardizing the *content* of health data, its scope is focused on *what* data elements should be exchanged for interoperability.[30] It does not inherently define the
  mechanisms for securing this data during transport or ensuring its continuous availability across networks. Cloudflare fills this critical gap by providing the necessary security and performance infrastructure for the data in transit, ensuring that the standardized USCDI data can be exchanged securely and efficiently. This includes robust encryption for data in transit [9], protection against cyber threats that could disrupt data flow [6], and a global network that ensures high availability and low latency for data exchange.[22]

TD-8: What are the most effective certification criteria for health IT to ensure interoperability and promote innovation?

Effective certification criteria for health IT should encompass robust security, privacy, and performance standards, alongside a commitment to accessibility. Cloudflare's adherence to a wide array of industry-recognized certifications demonstrates its commitment to these principles, which are foundational for interoperability and innovation:

- **Comprehensive Security Certifications:** Cloudflare maintains ISO 27001 (Information Security Management System), SOC 2 Type II (attesting to Security, Confidentiality, and Availability controls), and PCI DSS Level 1 certifications, providing independent assurance of its robust security controls.[31]
- **Privacy Information Management:** Cloudflare holds ISO 27701 (Privacy Information Management System) certification and aligns with GDPR requirements. Cloudflare will also certify to Global Cross-Border Privacy Rules (CBPR) and Global Privacy Recognition for Processors (PRP) systems in 2025, demonstrating a strong commitment to data privacy, which is critical for handling PHI.[31]
- **Government Compliance:** Cloudflare has achieved FedRAMP Moderate authorization, a critical certification for serving U.S. federal agencies, indicating its ability to meet stringent government security standards.[26]
- **Accessibility:** Cloudflare's dashboard completes a Voluntary Product Accessibility Template (VPAT) in compliance with Web Content Accessibility Guidelines (WCAG) 2.1 AA and Section 508 of the Rehabilitation Act, directly addressing accessibility requirements for digital health solutions.[26]

These comprehensive third-party certifications significantly reduce the due diligence burden for healthcare organizations and align with the RFI's implied need for trustworthy and inclusive digital health solutions.

TD-9: How should CMS/ASTP/ONC revise certification criteria to prioritize API-enabled capabilities and reduce the burden on health IT developers?

To prioritize API-enabled capabilities and reduce the burden on health IT developers, certification criteria should emphasize platforms and solutions that inherently simplify security and management, allowing developers to focus on innovation. Key revisions could include:

- **Prioritizing Integrated API Security:** Certification criteria should favor platforms that offer comprehensive, integrated API security capabilities. This includes features for automated discovery of API endpoints, robust protection against common API threats (such as those outlined in the OWASP Top 10 API Security risks), and built-in data leakage prevention. By embedding these security measures directly into the development and deployment environment, developers are freed from implementing complex security controls manually, significantly reducing their burden and accelerating the creation of secure API-enabled applications. [3]
- **Valuing Unified Management and Policy Enforcement:** Certification should recognize the benefits of platforms that provide a unified interface for managing applications and APIs. This approach reduces tool sprawl and alert fatigue, enabling developers and IT teams to enforce consistent security policies across their digital health products from a single control plane. This simplification of management and policy enforcement directly contributes to reducing operational burden and improving efficiency. [6]

TD-10: What are the challenges and opportunities in implementing the 21st Century Cures Act's API condition of certification for health IT developers?

Implementing the 21st Century Cures Act's API condition of certification presents both significant challenges and transformative opportunities for health IT developers:

Challenges:

- **Ensuring Robust Security for Sensitive Data:** The mandate to expose health information via APIs introduces new attack vectors. Developers face the complex challenge of securing these APIs against a wide range of cyber threats, including unauthorized access, data leakage, Distributed Denial-of-Service (DDoS) attacks, and sophisticated bot activity, all while handling highly sensitive Protected Health Information (PHI). [5]
- **Achieving High Performance and Scalability:** APIs must be designed to handle potentially massive volumes of requests and provide consistently low latency to support real-time health data exchange, clinical decision support, and seamless user experiences for patients and providers. Scaling these APIs efficiently and reliably under varying loads is a significant technical hurdle. [1]
- **Navigating Regulatory Compliance:** Adhering to stringent regulatory requirements, such as HIPAA and HITECH, for API data exchange adds substantial complexity. Developers must ensure meticulous attention to data privacy, implement robust security safeguards, maintain comprehensive audit trails, and potentially manage Business Associate Agreements (BAAs) with third-party services. [31]

Opportunities:

- **Accelerated Innovation in Digital Health:** The API condition of certification creates a clear and standardized pathway for developers to build innovative digital health products. By providing programmatic access to health data, it fosters the creation of new applications for health

management, care navigation, and personalized health guidance, ultimately benefiting Medicare beneficiaries. [1]

- **Enhanced Interoperability and Data Exchange:** The mandate promotes seamless health information exchange across disparate systems, breaking down data silos. This enhanced interoperability leads to improved care coordination, more informed clinical decision-making, and a more integrated and efficient health ecosystem. [1]
- **Reduced Information Blocking:** By requiring open, standards-based APIs, the Cures Act aims to reduce technical and practical barriers that contribute to information blocking. This fosters an environment where health information can flow more freely and securely, enabling better patient care and public health initiatives. [1]

Successfully navigating these challenges and capitalizing on these opportunities requires a strategic approach to API development and deployment, often leveraging robust, integrated platforms that can provide foundational security, performance, and compliance capabilities.

TD-11: What are the challenges and opportunities in implementing the EHI (Electronic Health Information) export capabilities for health IT developers?

Implementing Electronic Health Information (EHI) export capabilities presents both critical challenges related to data integrity and security, and significant opportunities to empower patients and enhance data portability.

Challenges:

- **Ensuring Data Security and Privacy:** The paramount challenge is to ensure that EHI is securely exported without unauthorized access, data leakage, or compromise of sensitive patient information. This requires robust data loss prevention capabilities that can detect and block the oversharing of sensitive data across various channels, including cloud applications and and devices. [19]
- **Maintaining Regulatory Compliance:** Health IT developers must navigate stringent regulatory requirements for EHI handling during export, including HIPAA. This necessitates platforms architected for data compliance, allowing consistent controls across different locations, users, and applications, and simplifying audit processes. [28]
- **Managing Performance for Bulk Export:** Large-scale EHI exports can be resource-intensive and time-consuming, potentially impacting system performance. Optimizing the delivery and transfer of large datasets efficiently is a key technical challenge. [23]

Opportunities:

- **Empowering Patients:** Secure and efficient EHI export capabilities directly empower patients with greater control and portability over their health data, aligning with the Cures Act's goals of patient access and engagement. [1]
- **Facilitating Data Portability and Interoperability:** By enabling robust EHI export features, health IT developers can support seamless data portability across different systems and organizations, fostering broader interoperability within the health ecosystem. This allows for more comprehensive patient records and supports continuity of care.

TD-12: What are the opportunities for CMS/ASTP/ONC to endorse or leverage non-CMS data sources (e.g., social determinants of health data, consumer-generated health data) to stimulate innovation?

Endorsing or leveraging non-CMS data sources offers substantial opportunities to stimulate innovation by providing a more holistic view of patient health. To realize these opportunities securely and compliantly, CMS/ASTP/ONC should consider solutions that facilitate:

- **Secure Ingestion and Access:** Platforms that offer secure ingestion and access to diverse data types, regardless of their origin, are crucial. This ensures the privacy and integrity of sensitive information from sources like social determinants of health (SDOH) or consumer-generated health data, allowing for extensible security controls across different systems. [2]
- **Unified Policy Enforcement:** A unified policy engine that allows organizations to define security rules once and apply them consistently across various data environments is vital. This ensures consistent data protection and security policies, even when incorporating new, non-CMS data sources, thereby simplifying governance. [2]
- **Data Localization and Sovereignty:** For sensitive consumer-generated health data, particularly from diverse populations, data localization services are essential. These services allow organizations to precisely control where traffic is inspected and where logs are sent, addressing data residency and sovereignty concerns crucial for building trust and ensuring compliance. [28]

By enabling the secure and compliant integration of these diverse data sources, CMS/ASTP/ONC can foster the development of innovative digital health products that address a wider range of patient needs and provide more comprehensive insights into health outcomes.

TD-13: What are the opportunities for technology vendors to leverage full EHI access to develop innovative digital health products?

Full Electronic Health Information (EHI) access presents immense opportunities for technology vendors to develop truly innovative digital health products. To confidently leverage this access, vendors require robust underlying security and performance infrastructure.

Opportunities:

- **Development of Advanced Analytics and AI Tools:** Full EHI access enables the creation of sophisticated analytics and artificial intelligence (AI) tools for predictive modeling, personalized treatment plans, and population health management, leading to more proactive and effective care. [16]
- **Creation of Comprehensive Patient Engagement Platforms:** Vendors can develop more integrated and personalized patient engagement platforms that offer a complete view of a patient's health, facilitating better self-management, care coordination, and shared decision-making.
- **Streamlined Clinical Workflows:** Access to comprehensive EHI allows for the development of tools that can significantly streamline clinical workflows, reduce administrative burden, and improve efficiency for healthcare providers. [1]

To realize these opportunities, the underlying infrastructure must provide:

- **Robust API Security:** Secure API solutions are crucial for protecting APIs that expose EHI, preventing data leakage, and ensuring only authorized access. This builds trust and confidence for vendors to integrate with EHI sources and develop new applications. [3]
- **Accelerated Application Performance:** Global network infrastructure ensures that digital health products leveraging EHI can deliver fast and seamless experiences to users, which is vital for adoption and engagement. [3]
- **Reduced Risk of Data Breaches:** Solutions that significantly reduce the likelihood and costs of data breaches de-risk the development of innovative products that rely on full EHI access, encouraging investment and creativity in the digital health space. [6]

TD-14: How are networks currently using FHIR APIs to facilitate health information exchange, and what are the challenges and opportunities?

Networks are increasingly using FHIR APIs as a cornerstone for standardized, real-time health information exchange, moving towards a more interoperable ecosystem.

Challenges:

- **API Security:** Protecting FHIR APIs from various cyber threats (e.g., OWASP Top 10, DDoS attacks, malicious bots, shadow APIs, data leakage) is a significant challenge. Integrated API security solutions are essential to address these vulnerabilities and ensure the integrity and confidentiality of exchanged data. [3]
- **Performance and Scalability:** Ensuring FHIR APIs can handle high volumes of requests and provide consistently low latency is critical for real-time data exchange and clinical decision support. Global network and content delivery capabilities are essential to maintain performance and availability, especially as data exchange scales. [3]
- **Authentication and Authorization:** Managing access to FHIR APIs for diverse users and systems (patients, providers, payers, applications) requires robust authentication and granular authorization mechanisms to ensure only legitimate, authorized clients can access sensitive data. [3]

Opportunities:

- **Enhanced Interoperability:** Secure and performant FHIR APIs enable seamless data flow between disparate systems, improving care coordination, reducing data silos, and supporting a more integrated patient-centric learning health system. [1]
- **New Digital Services and Workflows:** The reliable exchange of health information via FHIR APIs facilitates the development of new digital health services and applications that can leverage real-time data for improved outcomes, streamlined administrative processes, and innovative patient engagement tools.
- **Mandating Cloud-Based Security Platforms:** CMS/ASTP/ONC has an opportunity to encourage or mandate the use of robust, cloud-based security platforms to protect FHIR endpoints from attacks. Such platforms offer advanced threat intelligence, scalability, and specialized API security features that are critical for defending against the evolving landscape of cyber threats, thereby ensuring the resilience and trustworthiness of the entire FHIR-enabled ecosystem. [6]

TD-15: What are the challenges and opportunities in implementing bulk FHIR APIs for health information exchange?

Implementing bulk FHIR APIs for health information exchange presents specific challenges related to data volume and security, alongside significant opportunities for large-scale data synchronization and analytics.

Challenges:

- **Network Capacity and Performance:** Bulk FHIR API transfers can generate significant traffic, potentially overwhelming network infrastructure. Global network infrastructure, including content delivery networks (CDNs) and smart routing capabilities, is designed to handle massive data volumes efficiently, ensuring fast and reliable bulk transfers. [23]
- **DDoS Protection:** Large data transfers can be targets for Distributed Denial-of-Service (DDoS) attacks or inadvertently trigger rate limits, disrupting critical data flows. Comprehensive DDoS protection is essential to safeguard against volumetric threats, ensuring uninterrupted service for bulk API calls. [6]
- **Data Integrity and Security:** Maintaining the integrity and security of large datasets during bulk transfer is critical. Security features, including robust SSL/TLS encryption for data in transit and API security solutions, are necessary to protect data from compromise. [23]

Opportunities:

- **Efficient Data Synchronization:** Bulk FHIR APIs, supported by high-performance infrastructure, enable efficient synchronization of large health datasets. This is invaluable for applications in analytics, population health management, research, and data warehousing.
- **Scalable Data Exchange:** The implementation of bulk FHIR APIs allows for scalable data exchange, supporting the needs of large healthcare systems, health information exchanges (HIEs), and data providers for comprehensive data aggregation and distribution.

TD-16: What are the tradeoffs between point-to-point FHIR API connections and shared network infrastructure for health information exchange?

The choice between point-to-point FHIR API connections and a shared network infrastructure for health information exchange involves distinct tradeoffs, with shared infrastructure generally offering greater scalability, security, and efficiency.

Benefits of Shared Network Infrastructure (e.g., via a global cloud platform):

- **Centralized Security and Policy Enforcement:** A shared infrastructure that provides a unified control plane for security policies reduces the overall attack surface and ensures consistent protection across all participants, unlike disparate point-to-point connections that require individual security configurations. [6]
- **Improved Performance and Reliability:** Global network infrastructure optimizes traffic routing and content delivery, leading to better performance and higher availability for all connected entities compared to individual point-to-point setups, which can suffer from latency and single points of failure. [3]

- **Reduced Operational Overhead:** Managing security, performance, and compliance for numerous point-to-point connections can be complex and resource-intensive. A shared infrastructure simplifies these operations, reducing tool sprawl, alert fatigue, and the need for redundant security implementations. [6]
- **Enhanced Scalability and Ecosystem Growth:** Shared infrastructure can more easily scale to accommodate new participants and increasing data volumes without requiring individual reconfigurations for each new connection, fostering broader ecosystem growth and interoperability. [18]

Tradeoffs of Shared Network Infrastructure:

- **Initial Setup Complexity:** While modern platforms simplify deployment, the initial setup and integration of a comprehensive shared network infrastructure may require more coordination and upfront planning compared to establishing a simple point-to-point connection. However, the long-term benefits in security, performance, and manageability often outweigh this initial investment.

TD-17: What role should CMS/ASTP/ONC play in ensuring the viability and functionality of data sharing networks (e.g., QHINs)?

CMS/ASTP/ONC play a crucial role in ensuring the viability and functionality of data sharing networks like Qualified Health Information Networks (QHINs) by establishing clear standards, fostering trust, and encouraging the adoption of robust technical foundations. This role includes:

- **Setting and Enforcing Security Benchmarks:** CMS/ASTP/ONC should promote the adoption of comprehensive security frameworks, such as Zero Trust, and require adherence to independently audited certifications (e.g., ISO 27001, SOC 2 Type II, FedRAMP Moderate). This ensures a high baseline of security for all participants within data sharing networks, protecting sensitive health information. [6]
- **Ensuring Performance and Resilience Standards:** Encouraging the use of global network providers with high uptime Service Level Agreements (SLAs) and advanced traffic management capabilities is crucial. This ensures continuous availability and fast data exchange within data sharing networks, which is vital for real-time clinical workflows and patient access. [3]
- **Promoting Secure API Practices:** Mandating robust API security measures, such as those offered by integrated API security platforms, is essential to protect the critical data exchange points within these networks and build trust in the interoperability framework. [3]
- **Facilitating Auditability and Transparency:** Policies should encourage solutions that provide granular logging and analytics capabilities, enabling comprehensive audit trails of data access and exchange. This transparency is key for accountability and addressing information blocking concerns. [23]

By actively guiding and supporting the technical underpinnings of these networks, CMS/ASTP/ONC can ensure they are secure, reliable, and capable of fulfilling the vision of a truly interoperable health ecosystem.

TD-18: What information blocking practices are most prevalent among technology vendors, data providers, and networks, and how can they be mitigated?

While information blocking can stem from various factors, including business practices and legal interpretations, many prevalent practices have underlying technical barriers that can be mitigated through robust infrastructure.

Prevalent Information Blocking Practices (with Technical Mitigation Strategies):

- **Security Concerns Leading to Data Withholding:** A primary reason for reluctance to share data is the fear of data breaches, cyberattacks, or compliance violations. This can manifest as overly restrictive access policies.
  - **Mitigation:** Robust security solutions, including comprehensive DDoS protection, advanced bot management [25], and Web Application Firewalls (WAF) [10], create a secure environment where organizations can confidently open up data access via APIs without fear of compromise. This reduces the technical rationale for withholding data due to security anxieties.
- **Performance Issues Hindering Data Access:** Slow or unreliable data access can act as a de facto form of information blocking, frustrating users and hindering legitimate data retrieval.
  - **Mitigation:** Global network infrastructure, including Content Delivery Networks (CDNs) and smart routing, ensures that data access is fast and reliable. By optimizing data pathways and caching content closer to users, performance barriers to information flow are removed. [3]
- **Lack of Transparency and Auditability:** Concerns about tracking data flows, ensuring proper authorization, and demonstrating accountability can lead to reluctance in sharing, as organizations may fear inability to prove compliance.
  - **Mitigation:** Granular logging and analytics capabilities provide transparency and auditability for all data access and exchange events. This level of visibility helps address concerns related to information blocking complaints by providing clear evidence of data flow and access attempts, fostering trust and encouraging compliance with interoperability mandates. [23]

By addressing these technical impediments, the underlying infrastructure can enable the "seamless and secure flow of health information" [1] that is the antithesis of information blocking, thereby fostering an environment of trust and transparency.

TD-19: What are the shortcomings in the implementation of price transparency requirements for technology vendors, data providers, and networks, and how can they be addressed?

Shortcomings in the implementation of price transparency requirements often relate to the technical challenges of securely and reliably exposing large volumes of dynamic data. Addressing these requires a focus on robust digital infrastructure.

Shortcomings and How They Can Be Addressed (via Digital Infrastructure):

- **Inconsistent Data Availability and Accessibility:** Price transparency data, if not consistently available or easily accessible, fails to meet its objective. This can be due to unreliable systems or poor network performance.
  - **Address:** Global network infrastructure and performance optimizations ensure that price transparency information is consistently accessible to all stakeholders, regardless of their geographic location or network conditions. This includes leveraging content delivery networks to distribute data closer to users. [3]

- **Security Vulnerabilities in Data Exposure:** Exposing sensitive pricing data, even if not PHI, can still be a target for manipulation, denial-of-service attacks, or unauthorized access, undermining trust in the data.
  - **Address:** Robust API security and Web Application Firewall (WAF) solutions are crucial to ensure the integrity and availability of price transparency data. These solutions protect against various cyber threats, preventing manipulation, unauthorized access, or service disruption. [23]
- **Scalability Issues During Peak Demand:** If price transparency tools experience high traffic surges (e.g., during open enrollment periods or policy changes), underlying systems can become overloaded, leading to service interruptions.
  - **Address:** The ability to manage peak loads effectively, through solutions like virtual waiting rooms or advanced load balancing, ensures continuous availability. This prevents system overloads that could hinder access to critical information when it is most needed. [23]

By providing a secure, performant, and reliable foundation, digital infrastructure plays a critical role in ensuring that the technical implementation of price transparency requirements is robust and effective, thereby supporting CMS's goals for informed health decisions.

Conclusion

Cloudflare is committed to supporting CMS and ASTP/ONC in building a robust, secure, and interoperable "Health Technology Ecosystem." Our solutions directly address the challenges and opportunities outlined in Section E of the RFI, providing the foundational security, performance, and identity management capabilities necessary to stimulate innovation, enhance data exchange, and streamline workflows for all stakeholders. We believe our proven track record in the healthcare sector and our comprehensive, integrated platform make Cloudflare a strategic partner in realizing the vision of a truly patient-centric learning health system.

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